SPMM Smart Revise

Old Age Psychiatry

Paper B Syllabic content 8.1

1. Demographics of Old Age Psychiatry

A. Changes in the UK and worldwide

- Between 2000 and 2050, the proportion of the world's population over 60 years will double from about 11% to 22%
- The absolute number of people aged 60 years and over is expected to increase from 605 million to 2 billion over the same period
- 10 million people in the UK are over 65 years old (one sixth of the population) – projected to be 19 million by 2050 (one quarter of the population)
- 3 million in the UK are over 80 years old
- Growing numbers of elderly people are having an impact on the NHS - average spending for retired households is nearly double that for non-retired households

B. Service provision

- Older adults under the care of mental health services have complex presentations with problems in multiple domains – psychological, cognitive, functional, behavioural, physical and social
- **The National Service Framework for Older People (2001)** contained a key standard that older people with mental health problems should have access to specialist services
- **Everybody's Business (2005)** clarified that older people's mental health problems require input from both health and social care, physical and mental health services and mainstream and specialist services
- **No Health Without Mental Health (2011)** outlined an expectation that services be age appropriate and non discriminatory
- There has been a recent trend of specialised older adults services being reduced and a movement to ageless services in some areas of the UK. These changes have been highlighted as potentially breaching the **Equality Act 2010** by causing indirect discrimination and reducing patient choice. The need for specialist services has been outlined in commissioning guidance issued by the Joint Commissioning Panel for Mental Health Services (JCPMH). This guidance emphasizes the need for close working with social services, primary care, community services, voluntary services and geriatricians

Needs based criteria for Older People's Mental Health Services developed for commissioners by the Royal College of Psychiatrists

People of any age with a primary dementia

People with mental disorder and physical illness or frailty, which contribute(s) to, or complicate(s) the management of their mental illness. This may include people under 65

People with psychological or social difficulties related to the ageing process, or end of life issues, or who feel their needs may be best met by a service for older people. This would normally include people over the age of 70

C. Role of carers:

- Older people with mental health problems are likely to need greater carer support, often provided by family members who are also older
- 30% of carers will suffer depression at some stage

- Inability of carers to continue to provide support has been found to be a trigger for movement from home into long term care settings

10 key points from JCPMH Guidance for commissioners of older people's mental health services
1. Older people will form a larger proportion of the population 2. Older people's mental heath services in particular benefit from an integrated approach with social care services 3. Older people's mental health services need to work closely with primary care and community services 4. Services must be commissioned on the basis of need and not age alone 5. Older people's mental health services must address the needs of people with functional illnesses such as depression and psychosis as well as dementia 6. Older people often have a combination of mental and physical health problems 7. Older people's mental health services must be disciplinary 8. Older people with mental health needs should have access to community crisis or home treatment services 9. Older people with mental health needs respond well to psychological input 10. Older people should have dedicated liaison services in acute hospitals

D. Specialist aspects of assessment of mental health in older people.

- Assessment of older adults presenting with mental health problems needs to incorporate many different domains as follows -
- Cognitive assessment
- Assessment of functional abilities
- Assessment of physical health issues and awareness of how these can interact and impact on mental health issues
- Understanding of role of imaging in dementia diagnosis
- Assessment of carer needs and holistic approach to care
- Assessment of capacity issues and awareness of relevant legislation, particularly in regards to long term care, such as the Mental Capacity Act, and Deprivation of Liberty Safeguards (DOLS)
  - The initial assessment of an older person with a possible mental illness may take place in a variety of settings, e.g. at home, in a residential or nursing home, outpatients, A+E and general hospital wards
  - There are many advantages to undertaking new assessments at the patient's home, particularly for people with dementia – depth and quality of information, and avoidance of a potentially tiring and expensive journey for patients
  - Physical examination, cognitive examination and informant interview are particularly important elements of the assessment

2. Psychological aspects of physical disease

Parkinson's disease

- Medication regimens and timings are very important- failure to maintain normal dosing schedules can result in delirium and depression, slowed cognition and anxiety
- Depression occurs in approximately two thirds of patients with PD, and dementia in approximately 40%
- Common cognitive deficits in PD are higher executive dysfunction, attention, memory, visuomotor processing and visual attention

Cerebrovascular disease

- Delirium affects 30-40% of people in the week after a stroke
- Depending on the region affected, focal cognitive deficits may result
- Pathological crying and emotional lability are relatively common post stroke and can be treated with SSRIs

Sensory impairment

- Older people have a higher incidence of sensory impairment
- Visual impairment can lead to Charles Bonnet syndrome– visual hallucinations in the absence of psychotic symptoms
- Charles Bonnet syndrome is most commonly associated with macular degeneration, also associated with cataracts and diabetic retinopathy
- Auditory impairment has been associated with psychotic symptoms in the elderly

Emotional reaction to illness and to chronic ill health

- Reaction to illness is dependent on multiple factors, including premorbid personality, perceived threat of the illness, treatment required and experiences of treatment
- People who tend to be anxious may have worsening of their anxiety such that it becomes pathological as a result of physical illness, likely due to an increased focus on physical sensations and symptoms and a morbid interpretation of them
- Metabolic changes during illness can accentuate the emotional response to it – dehydration, electrolyte imbalance, endocrine changes and infection can all produce affective symptoms
- Adjustment disorders are common following physical illness and by their nature are transient
- Depression is approximately 2-3 times more common in people with a chronic physical health problem, and occurs in around 20% of people suffering from chronic physical illness. Choice of an antidepressant should take into account side effects, which may impact on the underlying physical illness - e.g. SSRIs may worsen hyponatraemia, or increase risk of bleeding – and interaction with other medication. NICE advise that there is no evidence supporting the use of specific antidepressants for patients with particular chronic physical health problems, and a generic SSRI should be first line

3. Dementia syndromes in the elderly

A. Potentially reversible causes of dementia

- Intracranial causes
  - Normal pressure hydrocephalus
  - Subdural Haematoma
  - Cerebral Tumours
  - General paralysis of the insane (tertiary syphilis)
- Systemic disorders
  - Alcoholism
  - Anoxia
  - Hypoglycaemia
  - Myxoedema
  - Vitamin deficiencies
  - Drug or chemical poisoning
  - Pseudodementia
  - Renal and hepatic disease

Alcohol related dementia: A relatively common cause of young onset dementia (YOD), accounting for 12% of cases. Heavy prolonged use can cause damage to limbic structures and frontal lobes, leading to memory and excutive impairments. Memory impairment may be static but can improve following a period of abstinence. Autobiographical memory is often affected. Confabulation can occur. Neuroimaging may be non-specific or may show generalised cortical atrophy with frontal preponderance

Normal pressure hydrocephalus

- NPH is a syndrome where there is dilatation of cerebral ventricles (especially 3rd ventricle) and normal CSF pressure at lumbar puncture. It typically presents with the triad of: dementia, gait ataxia, and urinary incontinence.
- The population prevalence in the elderly is observed to be 0.4% (Trenkwalder et al 1995). The diagnosis rests on clinical suspicion. Disturbance of balance and mildly broad based, symmetrical short stepped gait are cardinal features. Gait disturbance always precedes the development of other symptoms and worsens insidiously over months
- There is progressive slowing of cognitive and motor functioning consistent with a pattern of subcortical dementia (pronounced slowness of thought, difficulties in sustaining, switching attention and difficulties in planning). Dementia is potentially reversible if NPH is treated promptly.
- 50% cases are idiopathic, 50% are secondary to mechanical obstruction of CSF flow across the meninges due to infection, trauma, subarachnoid haemorrhage etc.
- Urinary incontinence is a late symptom. Urinary urgency, frequency and incontinence are common but non-specific features

- CT scan shows increased size of the lateral ventricles and thinning of the cortex.
- The most widely used investigation is CSF tap test, where 40-50 ml is withdrawn by lumbar puncture with assessment of gait and cognition before and afterwards – but low sensitivity and negative predictive value
- Usual treatment is surgical placement of a ventriculo-peritoneal shunt, with the best candidates for ventriculo-peritoneal shunt being those whose NPH is secondary to an identified cause
- Gait impairment is the feature most likely to improve after shunting. The milder the dementia, the better the chance of a good outcome

Chronic subdural haematoma (SDH)

- Subdural veins are more vulnerable to tears in older people due to cortical shrinking
- SDH should be suspected where there is a changing pattern in cognitive function, especially if risk factors for SDH exist: post trauma; elderly after a fall; elderly after a head injury, infancy, cerebral atrophy, alcoholism, epilepsy, clotting disorders, pre-disposing drugs such as aspirin, Warfarin etc
- In 30% of cases there is bilateral SDH
- A history of head injury occurs in only 50% of patients
- A SDH may only manifest with symptoms months after it develops, therefore there may be no history of recent trauma
- Common features include headache, drowsiness, altered level of consciousness, and confusion, often with fluctuations in severity, progressing to a picture similar to subcortical dementia
- CT scan shows crescent shaped haematoma compressing sulci and midline shift – which may only be seen after 3-4 weeks
- CT scan during the first 3 weeks may not show the SDH as the clot is isodense during the early phase
- Treatment may be surgical via burr holes, or conservative using steroids such as dexamethasone
- Recovery after surgery can be dramatic, but complications include seizures and re-bleeding
- Mortality is around 10% - highest mortality rates are in those with depressed consciousness level and bilateral haematomas

B. Secondary causes of dementia

Huntington's Disease Dementia

- Huntington's Disease is one of the commonest inherited neurodegenerative illnesses
- Caused by autosomal dominant, unstable expansion of a CAG nucleotide repeat on Huntingtin gene, chromosome
- Presents typically in 4th decade with frontal dementia and movement disorder
- Prominent deficits in attention, semantic verbal fluency, processing speed and executive function
- Recall is affected more than recognition suggesting problems with retrieval rather than encoding

Multiple Sclerosis

- Dementia is one of several cognitive and psychiatric disturbances seen in people with MS
- Key diagnostic test is MRI but note that distinguishing between demyelination and vascular damage can be difficult in older adults
- Diagnosis can be confirmed using CSF examination and evoked potentials

Other secondary causes of dementia include HIV Dementia, Wilson's Disease and limbic encephalitis

Prion diseases

Prion protein is a normal brain protein (PrP), coded for by the PRNP gene on chromosome 20, whose function is unknown. Prion related diseases occur when the protein undergoes changes, which render it insoluble. The diseases caused by prions are spongiform encephalopathies, which are transmissible dementias. There are four forms of the disease in humans, all of which are rare – Kuru, Creutzfeldt-Jakob Disease, Fatal familial insomnia and Gerstmann Straussler syndrome.

Sporadic CJD: The average worldwide prevalence is around 0.1 cases per 100 000 – CJD is the most common of the human prion diseases. The agent responsible for CJD is the pathological form of the prion protein PrPsc. The normal form of the protein is called **PrP^{C}**, while the infectious form is called **PrP^{Sc}**. The infectious form is resistant to proteases, which is an enzyme in the body that can normally break down proteins. The accumulation of infectious forms leads to rapid degenerative changes leading to severe atrophy in various parts of the brain.

- Most cases occur after the fifth decade, although onset can occur at any age
- The clinical picture is one of rapidly deteriorating dementia, myoclonus, cerebellar and extra pyramidal signs leading to death within a year
- Patients may present with non-specific symptoms such as lethargy, depression and fatigue
- Within weeks, more fulminant symptoms develop, including progressive cortical-pattern dementia, myoclonus and pyramidal and extra pyramidal signs
- Myoclonus becomes prominent as the disease progresses and patients may develop cortical blindness
- 85% cases are spontaneous or sporadic; 10% result from genetic mutation; 5% result from iatrogenic transmission during transplant surgery of dura, corneal grafts, and pituitary growth hormone
- CT shows some atrophy of cortex worse frontally and atrophy of cerebellum.

- ✶ MRI may show non-specific basal ganglia hyperintensities (high signal changes in the putamen and caudate head) – but only seen in a proportion of cases and are not included as part of diagnostic criteria
- ✶ EEG shows 'periodic complexes'; periodic bi or triphasic discharges against slight low voltage background. This characteristic periodic pattern is less frequently seen in genetic or human growth hormone related cases. It has <u>not</u> been seen in any case of variant CJD
- ✶ CSF proteins including 14-3-3 protein are often elevated. 4-3-3 is a normal neuronal protein and maybe released into the CSF in response to a variety of neuronal insults. It is therefore generally a non-specific finding and 14-3-3 analysis cannot be used as a general screening test for sporadic CJD
- ✶ The definitive diagnosis is made by post-mortem microscopic examination, which demonstrates spongiform neural degeneration and gliosis throughout cortical and subcortical grey matter, sparing the white matter tracts
- ✶ Treatment is symptomatic - Sodium Valproate and clonazepam may help to reduce the severity of movement disorders

<u>New variant CJD (vCJD):</u> The rise of vCJD followed an epidemic of bovine spongiform encephalopathy (BSE) in cattle. BSE is a prion disease of cows that is thought to have been caused by cattle feeds that contained CNS material from infected cows. The incubation period between the ingestion of contaminated meat and development of the disease is probably less than 20 years.

- ✶ The disease in humans affects mainly young men in their 20's and is characterised by early anxiety and depressive symptoms, followed by personality changes, and finally a progressive dementia
- ✶ Ataxia and myoclonus are prominent and the typical course is 1-2 yrs until death
- ✶ The 'pulvinar' sign is diagnostic and refers to symmetric high-signal-intensity changesaffecting the pulvinar and medial areas of the thalamus andthe tectal plate seen on FLAIR sequence MRI in > 70% of confirmed casesof variant CJD in which the patient has undergone fluid attenuated inversion recovery (FLAIR) sequence MRI
- ✶ CSF proteins including 14-3-3 protein are elevated but this is variable in the variant form
- ✶ Prion protein immunostaining is positive in lymphoid tissues, hence the diagnosis can also be made from a tonsillar biopsy (appears to be sensitive and specific)
- ✶ EEG shows no distinctive changes but slow waves may be noted diffusely.

C. Alzheimer's disease

> ***RISK OF ALZHEIMER'S IN RELATIVES***
>
> *An actual predicted risk of developing Alzheimer's disease in the first-degree relatives of probands with Alzheimer's disease is 15-19% (one in five and one in six), compared with 5% in controls.*
>
> *This translates to a risk of developing Alzheimer's disease that is increased some 3 – 4 times relative to the risk in controls.*

- ✶ Dementia of the Alzheimer's type is the most common dementing disorder reported in clinical and neuropathological prevalence studies from Europe, North America and Scandinavia
- ✶ Onset is usually between the ages of 40 and 90, most often after age 65;
- ✶ Risk of Alzheimer's dementia increases with age: 1% at age 60, 5% at age 65, doubles every 5 years, 40% of those aged 85.
- ✶ It becomes more common with increasing age; among persons older than 75 years, the risk is six times greater than the risk for vascular dementia. Age at onset is earlier in patients with a family history of the disease.
- ✶ In geriatric psychiatric samples, Alzheimer's dementia is a much more common etiology (50-70%) than vascular dementia (15-20%)
- ✶ Although it is commonly diagnosed in the clinical setting after other causes of dementia have been excluded, the final diagnosis of Alzheimer's disease requires a neuropathological examination of the brain

Risk Factors:

- ✶ Proven: Age, Down's syndrome, Apolipoprotein ε4 allele
- ✶ Likely: Female sex, Head injury, postmenopausal oestrogen decline
- ✶ Possible: Family history of Down's syndrome, Family history of Parkinson's disease and vascular factors

Protective factors:

- ✶ Proven: Apolipoprotein ε2 allele
- ✶ Possible: smoking, NSAIDs, oestrogen, premorbid intelligence and education

The genetics of Alzheimer's disease

- ✶ Presenilin 2 gene (Chromosome 1) – early onset
- ✶ Presenilin 1 gene (Chromosome 14) – early onset
- ✶ Beta Amyloid precursor protein gene (Chromosome 21)- the gene for Amyloid precursor protein is found on the long arm and is implicated in the early onset dementia. The gene for amyloid precursor protein is on the long arm of chromosome 21. The beta amyloid protein, the major constituent of senile plaques, is a 42-amino acid peptide that is a breakdown product of amyloid precursor protein. People with Down syndrome (trisomy 21) have three copies of the amyloid precursor protein gene. Another cause of excessive deposition of the beta amyloid protein is a mutation on codon 717 in the Amyloid precursor protein gene. Amyloid PET scanning is a relatively new imaging technique which can be used to confirm the diagnosis of Alzheimer's dementia, but is mainly used in research
- ✶ Apolipoprotein ε4 (Chromosome 19). Presence of ε4 alleles increases risk of late onset dementia of Alzheimer's type. People with one copy of the gene have Alzheimer's disease have three times the

incidence of Alzheimer's disease than no ε4 gene, and people with two ε4 genes have the disease eight times more frequently than do those with no ε4 gene. Genetic testing for presence of ε4 is not currently recommended as it is also present in people without dementia, and not found in all cases of dementia

Criteria for the diagnosis of probable Alzheimer's disease

- Dementia established by clinical examination and documented by the MMSE, Blessed dementia scale or similar examination and confirmed by neuropsychological tests.
- Features required for a diagnosis: Deficits in 2 or more areas of cognition; Progressive worsening memory and other cognitive functions; No disturbance of consciousness and absence of systemic disorders or other brain diseases that in and themselves could account for the progressive deficits in memory and cognition

Diagnostic procedures in Alzheimer's disease:

- CT is mainly used to exclude other treatable causes. Other indications would include seizures, features suggestive of normal pressure hydrocephalus. In CT scans, cortical atrophy especially over parietal and temporal lobes, dilatation of the third ventricle which correlates with cognitive impairment (most common picture)
- MRI: Reduced grey matter, hippocampus, Amygdala and temporal lobe volumes
- SPECT: Characteristic reduction in blood flow in temporal and parietal regions (SPECT could distinguish specific features of dementia of Alzheimer type at early stages and frontal lobe dementia)
- PET: Reduced blood flow and metabolism in temporal and parietal regions
- MRS: Abnormal synthesis of membrane phospholipids early in the disease
- Amyloid PET imaging – shows deposition of beta amyloid even in preclinical stages of dementia

Clinical features of Alzheimer's disease:

- Cognitive symptoms:
  - Memory-deficits of short-term memory followed by long-term memory deficit later. Amnesia universal and is mainly for recent events. Disorientation is common, especially for time.
  - Language: Expressive and receptive dysphasia, lexical anomia (word-finding difficulty)
  - Apraxia characterised by inability to perform coordinated learnt motor tasks
  - Agnosia-inability to recognise peripheral sensory stimulation and recognise parts of the body
  - Impaired visuospatial skills and impaired executive functions.
- Psychiatric symptoms: delusions (15%), auditory and visual hallucinations (10-15%), and depression requiring treatment in 20% of patients.
- Behavioural and psychiatric symptoms (BPSD) are common in dementia. Apathy (59.6%) and depression (58.5%) were the most common abnormalities, followed by irritability (44.6%), anxiety (44%) and agitation (41.5%). The most common behavioural disturbance requiring intervention includes wandering and aggression/ anger outbursts. Some patients exhibit sexual disinhibition, incontinence, excessive eating and searching behaviour.
- The average survival expectation for patients with dementia of the Alzheimer's type is 8 years.

* Progression of the disease involves increased agitation, frequent emotional outbursts, night pacing, poor sleep and wandering. In the terminal phase, patients become profoundly disoriented, amnestic and incontinent of urine and faeces.

Summary of Cognitive tests:

Test	Description	Areas of cognition tested
AMTS	Cut-off 7 or 8 /10, a few minutes to use but only covers memory and orientation. It is one of the simplest and longest established cognitive tests.It is used commonly in various settings such as hospitals, general practice and for subjects at home with some adaptations.It is used commonly in various settings such as hospitals, general practice and for subjects at home with some adaptations.	Memory and orientation only
MMSE	Cut off 24/30, takes 5-10 minutes to complete and is a standard baseline test. It is the most widely used cognitive test in Old age psychiatry and has been validated in a variety of populations. It is good for screening global cognitive dysfunctions as opposed to focal cognitive dysfunctions. However it is important to note that it is subject to variation with age, socio-economic status and educational achievement; moreover it is heavily weighted on verbal performance, which means that the performance of dysphasic patients is particularly poor.	Orientation, memory, concentration, language, praxis and gnosis
CAPE	(The Comprehensive Clifton assessment for the elderly) Intended to assess level of disability and estimate need for care.	
DRS	Clinical Dementia Rating Scale. The most commonly used scale to measure severity and stage the clinical illness ranging from 0 (none) to 0.5 (questionable dementia) through mild and moderate to severe dementia.	Rated in six domains: memory, orientation, judgment and problem solving, community affairs, home and hobbies and personal care.
ACE	Addenbrooke Cognitive Examination. 100 point scale, provides more detailed cognitive assessment	Rated across a number of domains –orientation, registration, recognition, recall, perceptual abilities, language, verbal fluency
NPI	(NPI; Cummings et al., 1994) Rates frequency and severity of a range of neuropsychiatric symptoms. The NPI-NH also rates occupational disruptiveness, a measure of caregiver distress.	It measures 12 behavioural areas (delusions, hallucinations, agitation, depression, anxiety, euphoria, apathy, disinhibition, irritability, aberrant
CAMCOG	It is a more comprehensive cognitive test, which covers a wide range of ability. It takes 40 minutes to complete. It gives a score out of 104.	Orientation, comprehension, perception, memory and abstract thinking
Clock Drawing Test	A revealing test of praxis, offers qualitative and quantitative information.	Praxis Higher executive function

(Ref: Seminars in Old Age Psychiatry Pg 11-15)

Antidementia drugs:

- Donepezil (Aricept), Rivastigmine (Exelon) and Galantamine (Reminyl) are cholinesterase inhibitors used to treat mild to moderate cognitive impairment in Alzheimer's disease. They reduce the inactivation of the neurotransmitter acetylcholine and, thus, potentiate the cholinergic neurotransmitter, which in turn produces a modest improvement in memory and goal-directed thought.
- Donepezil is well tolerated and widely used. It has a long plasma half-life of 70 hours permitting once daily dosing; almost total plasma protein binding; highly selective reversible inhibition of acetyl choline. The side effects of Donepezil are largely gastrointestinal including nausea, vomiting, diarrhoea and anorexia. Some patients develop headache, dizziness, syncope and muscle cramps.
- Rivastigmine and Galantamine appear more likely to cause gastrointestinal (GI) and neuropsychiatric adverse effects than does donepezil. None of these medications prevents the progressive neuronal degeneration of the disorder. Galantamine has direct nicotinic stimulatory action.
- Rivastigmine has been evaluated in the symptomatic treatment of patients with mild-to-moderate dementia associated with idiopathic Parkinson's disease. Rivastigmine appears to improve both cognition and activities of daily living in patients with PDD, resulting in a clinically meaningful benefit in a large number of cases. The most frequently reported adverse events with a higher rate in patients taking rivastigmine than placebo were nausea, vomiting and anorexia. Rivastigmine has an effect on Acetylcholinesterase and Butyrylcholinesterase. It has become available as a transdermal patch for a novel once daily administration.
- Memantine (Ebixa) protects neurons from excessive amounts of glutamate, which may be neurotoxic. Trials show benefits of memantine augmentation of donepezil. Cochrane review indicates use in DAT, vascular and mixed dementias. Memantine is a non competitive, PCP-site, NMDA antagonist that in theory, may be neuroprotective(may protect neurons from glutamate mediated excitotoxicity) and thus is termed a disease-modifying drug. It is better tolerated than the cholinesterase inhibitors. The most common side effects include dizziness, headache, fatigue, diarrhoea and gastric pain. Memantine is predominantly used in the treatment of moderate to severe Alzheimer's dementia when it is felt that cholinesterase inhibitors are not effective. Memantine is used in the treatment of mild to moderate Alzheimer's dementia in those who are unable to tolerate cholinesterase inhibitors or when they are contraindicated (e.g. severe cardiac conduction defects or severe asthma)
- A Cochrane Review in 2007 found that the evidence for benefit of Gingko biloba on cognition in individuals with dementia was not convincing. Several small, short-term randomized trials have had mixed results. Tacrine is rarely used, because of its potential for hepatotoxicity.

Anti-dementia drugs	Starting dose	Treatment dose
Donepezil	5 mg daily	10 mg daily
Rivastigmine	1.5 mg BD	6 mg BD
Galantamine	4 mg BD	12 mg BD
Memantine	5 mg daily	10 mg daily

CSM restriction on olanzapine and risperidone:

- ✶ Having reviewed the literature on the use of Olanzapine and Risperidone for the behavioural and psychological symptoms of dementia, the committee on safety of medicines (CSM) concluded that each was associated with at least two fold increase in the risk of stroke and should therefore be no longer used in dementia (Duff 2004)
- ✶ However Herrmann Et al (2004) found no difference in the risk of stroke between Olanzapine and Risperidone when compared with typical neuroleptics in the treatment of dementia

Poor prognostic factors in Alzheimer's disease:

- ✶ Being male
- ✶ Onset before 65 years
- ✶ Prominent behavioural problems
- ✶ Parietal lobe damage
- ✶ Observed depression
- ✶ Severe cognitive deficits such as apraxia
- ✶ Absence of misidentification syndrome

Psychosis in Alzheimer's disease:

- ✶ The prevalence of psychosis in people with Alzheimer's disease ranges between 30 and 50%. Delusions were more common than hallucinations (Bassiony et al 2000). Visual and auditory hallucinations are more common although hallucinations can occur in all modalities. (Tariot 1995). There is some evidence of the association of psychotic symptoms with a rapid decline in cognition in Alzheimer's disease. (Forstl et al 1994)
- ✶ Common types of misidentifying delusions seen in individuals with Alzheimer's disease
    - o The Capgras type: The false belief that previously known people (e.g. wife or other care givers) have been replaced by impostors
    - o The phantom boarder: A false belief that guests are living in the pseron's house
    - o The mirror sign: The individual identifies his or her own image as someone else
    - o The TV sign: Misidentification of television images as real
    - o The magazine sign: misidentification of magazine images as being real and existing in three dimensional space (Karim &Burns 2003)

D. Vascular dementia

- Vascular dementia is the second most common cause of dementia after Alzheimer's disease, accounting for 20% of cases.
- The presentation of vascular dementia is variable and the clinical spectrum is wide.
- The NINCDS-AIREN criteria requires evidence of cerebrovascular disease on both examination and on brain imaging and a relationship between the onset of dementia & cerebrovascular disease by a) dementia occurring within 3 months of a stroke or b) abrupt deterioration in cognitive function or fluctuating stepwise course.
- A median number of 4.5 neurological signs per patient were noted in a large cohort of vascular dementia patients. Reflex asymmetry was the most prevalent symptom (49%) irrespective of the extent of vascular insult. Measures of small vessel disease were associated with an increased prevalence of dysarthria, dysphagia, Parkinsonian gait disorder, rigidity, and hypokinesia and as well to hemimotor dysfunction. By contrast, in the presence of a cerebral infarct, aphasia, reflex asymmetry, hemianopia, hemimotor dysfunction, hemisensory dysfunction, and hemiplegic gait disorder were more often observed.
- Based on clinical features, it is divided into three main subtypes: Cognitive deficits following a single stroke, Multi-infarct dementia & Progressive small vessel disease (Binswanger's disease)
- ***Cognitive deficits following a single stroke***: Not all cases result in cognitive impairment; More often seen in midbrain and thalamic strokes; Cognitive deficits may remain fixed or recover either partially or fully.
- ***Multi-infarct dementia***: Multiple strokes lead to stepwise deterioration; Follows a number of minor ischaemic episodes; Risk factors for cardiovascular disease present and periods of relative stability is seen between strokes
- Progressive small vessel disease (***Binswanger's disease***): This is a subcortical dementia with a clinical course characterised by slow intellectual decline and generalised slowing. The clinical picture may be dominated by the dementia with slowness of thought, decreased short term memory, disorientation and concreteness. Motor problems like gait disturbance and dysarthria are common. Depression is not uncommon. Major clear-cut syndromes aphasia or apraxia tend to occur only in complicated Binswanger's disease. In advanced cases, pseudobulbar palsy may occur. Multiple microvascular infarcts of perforating vessels leads to progressive lacunae formation. MRI scan shows small distinct infarcts (Lacunae) or more generalised white matter hyperintensities (leukoariasis)

HACHINSKI SCORE INDEX

Abrupt onset (2)
Stepwise progression
Fluctuating course (2)
Normal confusion
Relative preservation of personality
Depression
Somatic complaints
Emotional incontinence
History of hypertension
History of strokes
Evidence of associated atherosclerosis
Focal neurological symptoms (2)
Focal neurological signs (2)

Unless marked, each item scores one point. Score < 4 unlikely, scores > 7 likely to be vascular dementia

Risk Factors for vascular dementia

- Old age, Hypertension (50% of patients giving a positive history), Ischaemic heart disease
- Smoking, Alcohol consumption
- Hyperlipidemia, Atrial fibrillation, Family history, Valvular disease, Atrial myxoma, Carotid artery disease
- APOE4 allele
- Polycythemia, Sickle cell anaemia, Coagulopathies

Haschinski Ischemic score index (HIS):
A scoring system was developed by Haschinski and colleagues, which allow the clinician to quantify the likelihood of a patient having vascular, rather than degenerative dementia. It has been widely used to differentiate Alzheimer's and vascular dementia to some extent. It is often used as a checklist to detect vascular risk factors.

Alzheimer's Disease	Vascular Dementia
Females more affected than males	More common in males
Onset – insidious	May be sudden onset
Gradual progressive course	Stepwise course
Focal neurological signs absent	Focal neurological signs present
Insight often lost	Insight often retained
Mood flattened or euphoric	Mood symptoms are uncommon
Somatic complaints uncommon	Somatic complaints reported frequently, eg dizziness and headaches
Vascular risk factors less likely to be present	Vascular risk factors more common

CADASIL (Cerebral Autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopthy): CADASIL is a form of vascular dementia. The disease is transmitted as an Autosomal dominant trait with high penetrance. Molecular genetic linkage studies have assigned the disease gene to the long arm of chromsome 19. Patients with this condition usually present with recurrent stroke around the age of 40-50 and there is often a history of migraine. Many patients later develop a subcortical dementia and pseudobulbar palsy. MRI shows widespread white matter changes

Imaging in vascular dementia:

- CT: Increased number of infarcts
- MRI: White matter lesions are more numerous and severe in Alzheimer's disease
- SPECT: Irregular perfusion deficits
- PET: Cerebral blood flow and metabolism reduced and uncoupled
- MRS: Absence of phospholipids changes allow differentiation from Alzheimer's disease

E. Dementia with Lewy Bodies

- ✶ Lewy Body Dementia (LBD) - accounts for 15-20% of cases of dementia in hospital and community based samples. LBD presents with progressive dementia with parkinsonism and a fluctuation in the level of attention and the severity of cognitive impairment.
- ✶ Lewy Bodies are eosinophilic intracytoplasmic neuronal inclusion bodies made up of abnormally phosphorylated neurofilament proteins, which are aggregated with ubiquitin and alpha-synuclein. These are scattered through the brainstem, subcortical nuclei, limbic cortex (Cingulate, entorhinal, Amygdala) and neocortex (frontal, temporal and parietal lobes)

Consensus criteria for the clinical diagnosis of LBD

- o **The central feature required for the diagnosis** is progressive cognitive decline of sufficient magnitude to interfere with normal social or occupational function.
- o **Core** features include two of the following, which are essential for a probable diagnosis, and one is essential for a possible diagnosis.
  - Fluctuating cognition with profound variations in attention and alertness
  - Recurrent visual hallucinations-that are typically well formed and detailed
  - Spontaneous motor features of parkinsonism (seen in 70% of cases)
- o **Supportive** features
  - Repeated falls due to autonomic dysfunction
  - Syncope
  - Transient disturbances of consciousness
  - Neuroleptic sensitivity
  - Systematized delusions
  - Hallucinations in other modalities

Clinical features:

- ✶ The prevalence rates of paranoid delusions and auditory hallucinations are 65% and 20% respectively. (McKeith et al 1996).
- ✶ Recurrent visual hallucinations (often people and animals) are seen in 60-80% of cases. These are typically well formed and detailed.
- ✶ People with LBD are extremely sensitive to antipsychotics (60% of cases) and even small doses can lead to worsening of parkinsonian symptoms
- ✶ About 50% of individuals experience life threatening adverse effects to antipsychotics (Mc Keith et al 1992) and severe reactions may be dose related.
- ✶ The mean survival rate of cognitive decline is similar to Alzheimer's disease but sudden deterioration over a period of 1-2 years can also occur
- ✶ There is a 10% decline rate per year in terms of worsening of parkinsonism.
- ✶ Increased frequency of e4 allele (APOE genotype) is seen in LBD as well.
- ✶ Some specific features of LBD include relative sparing of short-term memory. Deficits on test of attention and of frontal subcortical skills and visuospatial ability may be especially prominent. Relative sparing of medial temporal lobes on CT/MRI scan is also noted in LBD. Medial temporal lobes relatively preserved in SPECT-HMPAO scan

Pathological features:

- Lewy Bodies are essential for the diagnosis to be confiemd pathologically. Associated but not essential features include Lewy-Related neuritis, plaques of all morphological types, neurofibrillary tangles, regional neuronal loss-especially brainstem (locus cereleus and substantia nigra) and nucleus basalis of Meynert, synapse loss and microvacuolation (spongiform change)

Drug treatments in Lewy body dementia:

- Cholinesterase inhibitors have been shown to improve cognition, delusions and hallucinations in patients with LBD. Improvements in cognitive functioning are modest. A large multicenter double blind trial comparing Rivastigmine with placebo in patients with LBD showed significant improvements in delusions and hallucinations (Mc Keith et al 2000). Cholinesterase inhibitors are not currently licensed in the UK to treat LBD – some studies have shown that they may alleviate symptoms but a Cochrane review has questioned their benefit.

F. Dementia in Parkinson's disease:

- Parkinson's disease is the result of the degeneration of the subcortical structures, primarily the substantia nigra and also caudate, putamen and globus pallidus. Therefore it is a prototype of subcortical degenerative disease. Subcortical diseases including Parkinson's disease are thought to impinge on the three M's-movement, mood and mentation (cognition)
- Most patients complain of slowed thinking or bradyphrenia. The likelihood of cognitive symptoms is greater in those with late onset disease (after 70 years of age)
- 10% of patients develop go on to develop dementia every year. Aarsland and colleagues noted that over an 8 year follow up of patients with Parkinson's disease, 78% became demented.
- People with early Lewy body dementia and Parkinson's dementia have generally less severe visual and verbal memory deficits, but more marked executive dysfunctions such as planning, reasoning and sequencing than people with Alzheimer's disease. Other cognitive deficits such as apraxia, dysphasia. Alexia, agraphia, anomia and acalculia are also less pronounced.
- Some neuropsychological impairment has been demonstrated in most studies for even non-demented patients with Parkinson's disease. The impairments are primarily observed in visuospatial tasks and in the shifting of cognitive sets, as measured by the Wisconsin card sorting test or the Stroop test.
- Parkinson's disease dementia impairs quality of life, exacerbates care-giver's distress, increases the likelihood of residential care and doubles mortality (Burn and Mc Keith 2003). This is probably due to the dementia adding to the motor burden of Parkinson's disease by restricting treatment of motor symptoms.
- Most patients treated with Levodopa or dopamine agonists (Eg Apomorphine, Ropinirole etc) develop neuropsychiatric side effects such as visual hallucinations (most common), psychosis, anxiety, euphoria, mania, impulsive behaviour and delirium.
- Visual hallucinations with preserved insight are very common. Delusions often of persecution may also occur. The treatment of psychosis or hallucinations should involve an attempt at reducing the dose of Levodopa or dopamine agonists. When this strategy is ineffective, an atypical

antipsychotic medication should be used. Among them, clozapine is the most successful. However it is reasonable to try quetiapine before clozapine but the success rate may be low.

- Risk factors for developing psychosis and hallucinations in Parkinson's disease include older age, longer duration of illness, cognitive impairment or dementia, severity of the illness, sleep deprivation, the use of dopamine agonists and polypharmacy.
- Diagnostic issues: There is no consensus on whether the parkinsonism of Parkinson's disease is phenotypically different from Lewy Body dementia. A diagnosis of Lewy Body dementia is potentially applicable to people with Parkinson's disease who develop dementia. The criteria for the diagnosis of both conditions were established by an international workshop on the diagnosis of Lewy body dementia in 1996 (Mc Keith et al 1996).
    - **Lewy body dementia**: If both motor symptoms and cognitive symptoms develop within 12 months, then it is conventional to give a diagnosis of Lewy body dementia.
    - **Parkinson's disease dementia**: If the parkinsonian symptoms have existed for more than 12 months before dementia develops then a diagnosis of Parkinson's disease dementia is given
- SPECT can be used to help differentiate Parkinson's disease from Lewy body dementia with LBD scans showing a **greater caudate involvement** (Coloby & O'Brien 2004; Walker et al 2004)
- A SPECT study of blood flow has showed a similar pattern of deficits in Parkinson's disease dementia and Lewy body dementia with reduced perfusion of the precuneus and parietal cortex, a location thought to be associated with visual processing (Firbank et al 2003)
- DAT scan (A compound called 123I-FP-CIT) reflects dopamine deficiency that is reduced in individuals with Parkinson's disease and LBD. It is therefore widely used to differentiate LBD from Alzheimer's. Dopamine is. The DAT scan detects changes in the dopamine transporter responsible for allowing brain cells take up dopamine.
- Rivastigmine is licensed for the treatment of Parkinson's Disease Dementia

G. Frontotemporal Dementia

- FTD is a term that encompasses several forms of dementia and includes
  - Pick's disease - behavioural or frontal variant of FTD (most common form- see below)
  - Primary progressive aphasia
  - Semantic dementia
  - Corticobasal degeneration
- FTD is more likely to affect younger populations and the age of onset ranges between 40 and 75 years of age. It accounts for around 20% of presenile cases of dementia. The etiology remains unclear, but in some cases genetic linkage to chromosome 17 has been found.
- The early clinical features include prominent changes in personality (disinhibition, social misconduct, lack of insight) and behaviour (apathy, mutism and repetitive behaviours). As the illness progresses, symptoms of frontal and temporal dysfunction may become apparent, which includes behavioural rigidity, impulsivity, emotional lability, fatuousness, executive dysfunction and hyperorality.
- In contrast to Alzheimer's disease, memory is affected later and less severely. Spatial orientation is well preserved. Insight is characteristically lost early. It is characterised by asymmetrical focal atrophy of the frontotemporal regions. There is underlying neuronal loss, gliosis and subsequent spongiform change in the affected cortices.
- There is an association between Frontotemporal dementia and motor neuron disease. Up to 10% of patients with motor neuron disease show features of dementia and these individuals have an aggressive course of illness.
- Treatment - SSRIs are of limited benefit for behavioural symptoms. Acetyl Cholinesterase inhibitors are unlikely to be beneficial.

Diagnostic investigations:

- Bilateral asymmetrical abnormalities of the frontal and temporal lobes is seen in CT/MRI scan especially early in the disease process.
- SPECT - regional cerebral blood flow studies have demonstrated disproportionate decrease in blood flow, radio tracer uptake and glucose metabolism in the frontal lobe in patients with suspected frontal lobe dementia.
- Neuropsychology - impaired frontal lobe function such as deficiency in abstract thinking, attentional shifting or set formation and relatively spared memory, speech, and perceptuospatial functions.

Dementia in Pick's disease:

- Onset is slow with steady deterioration. Predominance of frontal lobe involvement is evidenced by two or more of the following: Emotional blunting; Coarsening of social behaviour; Disinhibition; Apathy or restlessness; Aphasia (non-fluent aphasia).

- ✶ In the early stages, memory and parietal lobe functions are relatively preserved. Early symptom would include personality change and social disinhibition preceding memory or other cognitive impairment
- ✶ The onset is usually between 45-65 years. Men are more affected than women. The average duration of illness is 8 years. There is positive family history in 50% of cases.
- ✶ It is caused by Autosomal dominant mutation in the Tau gene (chromosome 17q 21-22) with complete penetration.
- ✶ Pick's cells are pathognomic in Pick's disease and they appear swollen and stain pink on H and E stains. Demyelination and Fibrous gliosis of the frontal lobe white matter may also be found. Senile plaques and Neurofibrillary tangles are absent.
- ✶ CT/MRI- Mild generalised atrophy but marked atrophy of frontal and temporal lobes with sparing of the posterior third of the superior temporal gyrus called 'Knife blade atrophy'.

Primary progressive aphasia:

- ✶ Patients have a progressive decline in their language with a relative sparing of other cognitive deficits. Speech is usually non-fluent and effortful, output is usually poor and the patient may become mute in the later stages. Behavioural features may occur later in the disease. MRI scan shows predominant atrophy of the perisylvian region

Semantic Dementia:

- ✶ In contrast with primary progressive aphasia speech is fluent. Patients have a variety of language difficulties such as impaired understanding of word meaning, naming difficulties and use of substitute words. Other cognitive domains are preserved - behavioural features may occur later in the disease.
- ✶ MRI shows disproportionate asymmetric atrophy of the temporal lobe (of the left more than the right). The atrophy of the anterior temporal lobe is more pronounced than the posterior temporal lobe.

H. Early onset dementia

- Most patients who develop dementia before 65 years of age have dementia of the Alzheimer's type (AD). Others are likely to have vascular dementia, frontotemporal dementia, head injury, alcohol intoxication, metabolic disorders.
- Certain conditions like progressive supranuclear palsy, and corticobasal degeneration, frontotemporal lobar degeneration are rarely seen in patients of senile age.
- The etiology and factors responsible for the accelerated onset of the illness are not fully known
- Genetic abnormalities appear to be important in some types of presenile dementia, such as frontotemporal dementia with parkinsonism linked to chromosome 17 and familial Alzhiemers disease with early onset type.
- Rapid progression of cognitive impairment with neuropsychological syndromes and neurological symptoms has been considered a characteristic of early onset AD
- Language problems and visuospatial dysfunctions are common in early onset AD.
- There are at least three dominant genes that have been identified in cases of familial Alzheimer's disease with early onset, namely 1) the amyloid precursor gene (APP), 2) the genes encoding presenilin 1 (PSEN1) and 3) presenilin 2 (PSEN2).

(Ref: Psychogeriatrics. 2009 Jun;9(2):67-72. What is 'early onset dementia'?-Miyoshi K1)

Progressive Supranuclear Palsy (from Adam & Victor 8th edn p. 936)

- It is a differential diagnosis of LBD. Onset is in the sixth decade (range 45 to 75 years). Presents with difficulty in balance, abrupt falls, slurred speech, dysphagia, and vague changes in personality, sometimes with an apprehensiveness and fretfulness suggestive of an agitated depression.
- The most common early complaint is unsteadiness of gait and unexplained falling (retropulsion).
- It may take a year or longer for the characteristic syndrome—comprising supranuclear ophthalmoplegia, pseudobulbar palsy, and axial dystonia to develop fully. Difficulty in voluntary vertical movement of the eyes (vertical gaze palsy)
- Bell's phenomenon (reflexive upturning of eyes on forced closure of the eyelids) and the ability to converge are also lost eventually, and the pupils then become small.
- The upper eyelids may be retracted, and the wide-eyed, unblinking stare, imparting an expression of perpetual surprise, is characteristic.
- A proportion of patients do not demonstrate these eye signs for a year or more after the onset of the illness. A tendency to extension may lead to falling backwards. Subcortical dementia is characteristic.

4. Delirium

- ✶ Onset of clinical features is rapid with fluctuations in severity over minutes and hours (even back to apparent normality). Clouding of consciousness is seen along with reduced attention span and distractibility. Global impairment of cognition with disorientation, and impairment of recent memory and abstract thinking are typical features.
- ✶ Disturbance in sleep/wake cycle with nocturnal worsening of symptoms can occur along with psychomotor agitation and emotional lability. Perceptual distortions, illusions, and hallucinations-characteristically visual – are often reported. Speech may be rambling, incoherent, and thought disordered. There may be poorly developed paranoid delusions.
- ✶ Liposwski (1983) reported no detectable cause for delirium in between 5 and 20% of cases.
- ✶ 2 clinical subtypes of delirium are recognized.
- ✶ ***Hyperactive delirium***: characterised by increased motor activity, agitation, hallucinations and inappropriate behaviour
- ✶ ***Hypoactive delirium***: characterized by reduced motor activity and lethargy and has a poorer prognosis.
- ✶ ***Prevalence***: Among the elderly 10-15% of patients have delirium on admission and a further 10-40% develops delirium during the course of their hospital stay. Point prevalence varies across clinical populations. In the general population, it is 0.4%; General hospital admissions-9-30%; Post operative patients: 5-75%; Intensive care unit patients: 12-50%; Nursing home residents up to 60%
- ✶ Delirium usually has a sudden onset, usually lasts less than 1 week, and resolves quickly. It may last longer in elderly patients. There is often patchy amnesia for the period of delirium. It may be a marker for the subsequent development of dementia.
- ✶ The major pathway implicated is the dorsal tegmental pathway, which projects from the mesenchephalic reticular formation to the tectum and thalamus. The reticular formation of the brain stem is the principal area regulating attention and arousal. The major neurotransmitter hypothesised to be involved is acetylcholine.
- ✶ The EEG characteristically shows a generalized slowing of activity. It generally causes a diffuse slowing of the brain activity on the EEG, which may be helpful in differentiating delirium from depression and psychosis.

Feature	Dementia	Delirium
Onset	Slow	Rapid
Duration	Months to years	Hours to weeks
Attention	Preserved	Fluctuates
Memory	Impaired remote memory	Impaired recent and immediate memory
Speech	Word finding difficulty	Incoherent (slow or rapid)
Sleep/wake cycle	Fragmented sleep	Frequent disruption, day/night reversal
Thoughts	Impoverished	Disorganised
Awareness	Unchanged	Reduced
Alertness	Usually normal	Hypervigilant or reduced vigilance

(Adapted from Lipowski ZJ. *Delirium: Acute Confusional States*. Oxford: Oxford University Press; 1990)

Rating scales in delirium:

Rating Scale	Features
DRS (Delirium Rating Scale)	Most widely used to date Requires interpretation by a skilled clinician or information from multiple clinical sources Advantage of distinguishing delirium from dementia
MMSE (Mini Mental State Examination)	Emphasizes neuropsychological functions linked to left cerebral hemispheric activity Many of the core disturbances of delirium reflect non-dominant hemispheric functions, e.g. attention is related to non-dominant orbitofrontal, prefrontal and posterior parietal regions.
CTD (Cognitive Test for Delirium)	Newly developed instrument, allows detailed investigation of a range of neuropsychological functions e.g. orientation, comprehension, attention and memory Suitable for use in patients whose ability to interact with the examiner may be compromised by immobility, intubation or absence of verbal abilities.
CAM (Confusion assessment method)	High sensitivity and specificity Allows a diagnosis of delirium and is readily incorporated in routine clinical settings Reduced sensitivity when used by nursing staff ratherthan physicians

(From Meagher & MacDonald, Advances in psychiatric treatment: 2001)

Management of Delirium:

- Identify and treat precipitating cause. Provide environmental and supportive measures (education, reorientation, reassurance, adequate lighting, reduce unnecessary noise, consistent staffing).
- Avoid sedation unless severely agitated or necessary to minimise risk to patient or to facilitate investigation/treatment.
- Regular clinical review and follow-up (MMSE useful in monitoring cognitive improvement at follow-up.
- Correct sensory impairments (e.g. hearing aids; glasses). Optimise patient's condition and attention to hydration, nutrition, elimination, and pain control.
- Make environment safe (remove objects with which patient could harm self or others).
- The available evidence suggests that an antipsychotic, such as low dose Haloperidol is effective in alleviating a range of delirious symptoms in both hyperactive (agitated and restless patients) or hypoactive states (drowsy & lethargic) and is appropriate for most patients who require drug treatment

- NICE guidelines advocate short-term (< 1 week) use of haloperidol or olanzapine if conservative measures fail. Benzodiazepine use in delirium may increase agitation, and use in elderly patients increases risk of falls and disinhibition. Benzodiazepines may be particularly helpful where the delirium is caused by withdrawal of alcohol or sedatives.

5. Non-degenerative Psychiatric Disorders

A. Depression

Prevalence

- In the population aged over 65 the prevalence of clinically significant depression is 10-15% (considered worthy of treatment by a psychiatrist)
- The overall prevalence of major depression is estimated at 2-3%. This figure is increased in hospital subpopulations and higher still in residential homes (Fountaulakis et al 2003) [Community-0.5-1.5%; Clinical outpatients 5-10%; Clinical Inpatients 10-15%; Residential and nursing homes-15-30%]
- Depressive disorders are at least 2 to 3 times more common in hospitalized patients, nursing home residents, or outpatients with chronic medical disorders
- The 3 *C*'s—cardiovascular disease, central nervous system disorders (e.g. strokes, dementia, Parkinson disease), and cancer—are medical conditions associated with a high risk for depression.
- People with dementia- 25%. A quarter of people with dementia will develop depressive illness at some point during the course of their illness.
- Elderly African Americans have been noted to have less depression than elderly Caucasians and it could possible be because they stay more engaged within their communities than elderly Caucasians

Clinical features

- Low mood may be less prominent in depression affecting patients in old age (Gurland 1976) . Elderly people with new or recurrent depression are more **hypochondriacal** and more **delusional** than younger people (Gurland 1976). Older people report experiencing less negative emotions such as sadness, fear and anger than younger adults
- Compared to early-onset depression seen in working age adults, the symptoms of late-life depression are somewhat altered. The changes reported include
  - Reduced complaints of 'feeling sad'
  - More frequent reports of hypochondriasis and somatic concerns instead of sadness
  - Poor subjective memory - a dementia like picture
  - Late onset neurotic symptoms (Marked anxiety, obsessive-compulsive or hysterical symptoms)
  - Apathy and poor motivation may predominate
  - Further certain symptoms such as anorexia, weight loss and anergia can be hard to interpret and attribute due to the physical fraility of many elderly patients (Koenig et al 1997)

Early onset Depression	Later onset depression
Depressed cognitions including suicidal thoughts, thoughts of worthlessness (Reinhard et al 2000) Anxiety/neuroticism (Baldwin 1995) Higher rates of familial depression (Maier et al 1991) Psychological vulnerability (Vandenberg et al 2001) Greater familial morbidity for depression, alcoholism and sociopathy (Mandlewicz and Baron 1981)	Cognitive impairment detected in 70% of cases Psychomotor changes (Hickie et al 2001)-severe psychomotor retardation or agitation seen in up to 30% of depressed elderly patients Depressive delusions regarding poverty, physical illness or nihilistic in nature Paranoia common, auditory hallucinations may occur in severe depression (derogatory and obscene) Weight loss (Janssen et al 2006) Severe life stress (Vandenberg et al 2001) Frequency and severity of life events (physical illness, loss of a spouse) may be greater in later life than in the general population (Hughes et al 1988) Lifetime depressive symptoms and somatic symptoms as preclinical markers (Hein et al 2003)

- Risk factors for late-life depression include female sex, poor health, disability and poor perceived social support.
- Neuroimaging studies in late onset depression includes ischemic changes (Baldwin and Tomenson 1995), reduction in grey matter volume in frontal and temporal lobes, sulcal widening and reduction in the volume of the caudate nucleus (Krishnan 1991), ventricular enlargement (Dahabra et al 1998), and reduction in the volume of the hippocampus (Bell-McGinty et al 2002).
- Other findings include CT - Cortical atrophy and ventricular enlargement; MRI: Atrophy, ventricular enlargement, lesions in basal ganglia and white matter; SPECT - reduced cerebral blood flow, sparing the posterior parietal cortex.

Treatment approaches

- ***Antidepressants***: The response rate to antidepressants in older people with is comparable to younger adults. The number needed to treat for major depression treated with antidepressants is about four and is similar to other age groups (Chew-Graham et al, 2004). Although elderly individuals should be started on lower doses, treatment doses should be similar to those used in young adults. First line recommended treatment for late onset

STARTING ANTIDEPRESSANTS IN THE ELDERLY

Low starting dose
Gradual increase in dose
Prolonged trial periods (2-3 months)
Long maintenance period (up to 2 years; may be life-long)

depression is SSRI due to reduced side effects and relative safety in overdose (NICE recommendation). There is some increaser risk of gastrointestinal haemorrhage among older patients on SSRIs. All SSRIs can induce hyponatraemia. Older patients take longer to recover from depression and may take 6-8 weeks to respond to antidepressants. At least 30% of elderly patients with depression do not respond to antidepressenat medication. (Mottram et al 1998).

- ***ECT*** remains the most effective treatment available for severe depression, with a recovery rate in the region of 80%. It is well tolerated, even by very elderly people (Tew et al 1999). There is evidence that it is particularly effective in psychotic depression (Baldwin et al 2002). Elderly patients are more likely to suffer from post ECT confusion and cognitive impairment and therefore this should be closely monitored during treatment. Memory impairment is often worse with bilateral electrode placement although the response to bilateral treatment may be more rapid.
- ***Psychological Interventions:*** Older patients with depression are rarely offered a psychological intervention. There is emerging evidence that for older adults with mild to moderate depressive episodes, a psychological intervention is as effective as medication (McCusker at al 1998, Pinquart and Sorenson 2001). In major depression, a combination of antidepressants with psychotherapy is more effective than either of these treatments alone, especially in relapse prevention (Reynolds et al 1999). CBT is the best-established treatment in depression and good evidence exists for its effectiveness in older adults (Thompson et al 2001). Interpersonal therapy is also effective in Relapse prevention (Reynolds at al 1999). There is smaller but developing evidence for problem solving treatment (Arean at al 1993). Family therapy has been successfully adapted for use with older adults, including those with depression (Benbow et al 1990).

Prognosis

- Older adults are thought to be at greater risk for chronicity of depression than younger persons. But this has been recently challenged. With control for confounding variables, remission rates of depression inpatients in late life are little different from those in midlife, but relapse rates appear higher (Mitchell & Subramaniam, 2005). Mortality is higher in older patients with depression because of concurrent physical disorders (Tuma 2000). A meta-analysis of outcomes in depressed older adults estimated that at 2-year follow-up, 33% of subjects were well, 33% remained

Good prognostic factors	Poor prognostic factors
Onset less than 70 years old Short duration of illness Absent physical illness Good previous adjustment Good previous recovery	Severe life events during follow up period Poor medication adherence Severity of initial illness Co-morbid physical illness Presence of psychotic symptoms Duration of illness for more than 2 years 3 or more previous episodes Previous history of Dysthymia Cerebrovascular disease (including vascular depression)

depressed, and 21% had died.

- Depression exacerbates the poor outcome of medical illnesses. Elderly individuals with depression were almost four times more likely than those without depression to die within 4 months of a myocardial infarction. Platelet aggregation is raised in this group of patients; depression may increase the risk for cardiovascular disease. Elderly people with depressive symptomatology have poor T-cell responses to mitogens and high concentrations of plasma interleukin 6, which is indicative of inflammatory activity that might increase the risk for bone resorption, predisposing to fractures.

Summary of Depression scales

Scale	Features
Geriatric Depression Scale	15-items, 4/5 minutes to complete. Avoids somatic questions and so good for older patients. Score > 5 suggests depressive illness.
BASDEC (Brief assessment schedule depression cards)	Initially designed for use in liaison psychiatry, particularly useful with deaf subjects. Consists of a series of statements in large print on cards, which are shown to the patient, one at a time and answers 'true' or 'false'
Hamilton rating scale	It has a number of somatic items, which render it less appropriate for older subjects. It is a general adult scale that quantifies depression but is not a diagnostic tool.
MADRS (Montgomery-Asberg Depression Rating Scale)	It is sensitive to change in depressive illness and is not reliably answered by patients with dementia
Depressive Sign Scale	Nine items to help detect depression in people with dementia
CSDD (Cornell Scale for Depression in Dementia)	Best validated scale for detecting depression in dementing patients (better in mild/moderate than severe). Interviewer-administered, uses information both from the patient and an informant. Scale factor analysis reveals 4 to 5 factors, including general depression, biologic rhythm disturbances, agitation/psychosis, and negative symptoms.
Patient health questionnaire PHQ-9	Nine item self-report scale widely used in UK primary care. Easy to use and has demonstrated sensitivity to change. Probably less validated among older subjects.

(Ref: Seminars in Old Age Psychiatry Pg 11-15)

Cognitive Impairment in Depression

- Although depression in all ages is associated with some degree of impaired concentration and subjective difficulties with memory, these cognitive deficits seem to occur more when first onset is in older age (Holroyd and Duryee 1997). Studies also found that individuals with late onset depression had specific deficits in attention and executive functions (Consistent with frontal lobe dysfunction) whereas those with recurrent early onset depression exhibited deficits in episodic memory (Rapp et al 2005) which is consistent with temporal lobe dysfunction
- Some elderly patients develop dementia during episodes of depression that subsides after remission of depression (pseudodementia). Most of these patients have late-onset major

depression. A significant proportion of those with pseudo dementia are left with some form of cognitive impairment after remission of depression, and nearly 40% develop true dementia within 3 years of follow up (Alexopoulos, 2005).

- Reversible dementia may be an early manifestation of dementing disorder and close follow-up is needed.

Pseudodementia	Dementia
Onset can be dated with some precision	Onset can be dated only within broad limits
Symptoms of short duration before medical help is sought	Symptoms usually of long duration before medical help is sought
Rapid progression of symptoms after onset	Slow progression of symptoms throughout course
History of previous psychiatric dysfunction common	History of previous psychiatric dysfunction unusual
Patients usually complain much of cognitive loss	Patients usually complain little of cognitive loss
Patients emphasize disability and communicate strong sense of despair	Patients conceal disability and often appear unconcerned.
Nocturnal accentuation of dysfunction uncommon	Nocturnal accentuation of dysfunction common
Attention and concentration often well preserved	Attention and concentration usually faulty
Don't know answers typical	Near-miss answers frequent
Memory loss for recent and remote events Usually severe	Memory loss for recent events usually more severe than for remote events

Vascular depression

- Alexopoulous suggested that a distinct subtype of geriatric depression called vascular depression exist. He proposed that cerebral Ischemic damage to the frontal sub cortical circuits could predispose, precipitate and perpetuate late onset depression. Vascular risk factors were found to be highly significantly associated with late onset depression, but not with just increasing age. Vascular depression is a term describing the hypothesis that in old age, depression may be primarily caused by small vessel cerebrovascular disease.
- The clinical feature of this proposed new subtype would include
  - Apathy

- Psychomotor Retardation
- Poor executive function on cognitive testing
- Less depressive thinking such as guilt and worthlessness
- Late age of onset

* Affected individuals have more apathy, retardation, and lack of insight, and less agitation and guilt than do elderly individuals who are depressed without vascular risk factors. Verbal fluency and object naming are the most impaired cognitive functions in patients with this form of depression.
* Theories explaining the association between depression and vascular disease;
  - Increased platelet aggregation
  - Both depression and ischaemia may be secondary to atherosclerosis (Baldwin and O'Brien 2002)
  - Recurrent depression across the life span may increase the risk of vascular pathology (Baldwin and O'Brien 2002)
  - Damage to end arteries supplying sub cortical striato-pallido-thalamo-cotical pathways may disrupt the neurotransmitter circuitry involved in mood regulation, causing or predisposing to depression.
* MRI deep white matter lesions (DWMLs) are more common in depressed than non-depressed older people. It is more common in late onset than early onset depression. DWMLs predict a poorer response to treatment of depression.
* Simpson et al studied the association between sub cortical lesions and antidepressant response in late life depression, concluding that a poor response could be expected in patients with vascular depression, but patients may recover with ECT, although with an increased risk of post-treatment delirium. Drugs used for the prevention of cerebrovascular disease might, for example, reduce the risk for vascular depression. Antidepressants that promote ischaemic recovery—e.g. dopamine or norepinephrine enhancing agents—might be favoured in vascular depression and antidepressants that inhibit ischaemic recovery—e.g. adrenergic blocking agents—are best avoided (Alexopoulos, 2005).

B. Bipolar disorder

- ✶ Mania accounts for 5-10% of mood disorders in the elderly. The 1-year prevalence of BD among adults aged 65 and older is 0.4%, significantly lower than in younger adults (1.4%). Average age at onset is 55 years and female to male ratio is 2:1
- ✶ Mania presents a similar clinical picture as in younger patients. But it is more often followed by a depressive episode in older patients and mixed affective presentation seems more common.
- ✶ First episode mania in late-life is uncommon; but these patients have lesser familial loading than younger bipolar patients and have more secondary mania than bipolar disorder.
- ✶ Diagnosis is based on atleast one week of elated mood, over activity and self-important ideas. In severe cases grandiose delusions and hallucinations may be seen. Insight is almost always absent.
- ✶ Patients with first-episode mania in late life are twice as likely to have a comorbid neurological disorder. This may or may not explain the affective presentation. Especially, the phenomenon of bipolarity after many years of unipolar depression has led to speculation that cerebral organic factors may play a part in the aetiology of late onset mania. In support of this, cognitive function is significantly impaired in between a fifth and third of elderly maniacs. Furthermore studies have shown a high rate of cerebral white matter lesions in late life mania.
- ✶ The term secondary mania denotes manic illness that starts without a prior history of affective disorder in close temporal relationship to a physical illness or drug treatment and often in the absence of a family illness of affective illness. A large number of conditions have been associated with secondary mania including stroke (commonest cause and particularly right sided lesions), head injury, tumours, endocrine infections, HIV infection, medication including steroids and anti-parkinsonian drugs.
- ✶ Lithium is used as first line prophylaxis but usually lower dosages are indicated. A lower therapeutic range around 0.4 to 0.6mmol/L is suggested for prophylaxis (Shulman, 2002)
- ✶ Valproate is also a popular choice. In more severe illness, initiate antipsychotic treatment. Age appropriate doses of neuroleptics must be used.

C. Late-life psychosis

- ✶ Psychotic symptoms of acute onset are usually seen in delirium secondary to a medical condition, drug misuse and drug-induced psychosis. Chronic and persistent psychotic symptoms may be due to a primary psychotic disorders such as chronic schizophrenia, late-onset schizophrenia, delusional disorders or affective disorders (especially depression). Psychosis may also be due to neurodegenerative disorders, such as Alzheimer's disease, vascular dementia, dementia with Lewy bodies or Parkinson's disease or other chronic medical conditions.
- ✶ Kraepelin introduced the term '**paraphrenia**' in 1913. Around the same time a systematic description of paranoid features with onset in later life was published under the title *Involutional Paranoia* (Kleist, 1913). Ever since, two conflicting views exist with regard to late life psychosis termed widely as paraphrenia:
  - o Late paraphrenia is nothing more than the expression of schizophrenia in the elderly

    - Late paraphrenia is different from schizophrenia and is associated with a different set of pathogenic factors seen in the elderly.
- According to current consensus the late onset psychotic illness is subdivided into late onset (onset after 40 years of age) and very late onset (onset after 60 years of age).
- People with late paraphrenia represent approximately 10% of the elderly population of psychiatric hospitals. The reported prevalence of the disorder among the elderly living in the community ranges from 0.1 to 4% - incidence has been estimated to be between 10-26 per 100000 per year. The point prevalence of paranoid ideation in the general elderly population has been estimated to be 4%–6% (35–37), but most of these patients will have dementia. Significantly higher number of females are affected than males.

Clinical Features:

- Persecutory delusions are the most common symptoms of late paraphrenia; they are found in around 90% of patients (Almeida *et al*, 1995*a*). Auditory hallucinations occur in approximately 75% of cases. Visual hallucinations are observed in up to 60% of patients. First rank symptoms are less common while negative symptoms and thought disorder are extremely uncommon. Few patients may present with delusions only (10-20%). Partition delusions (attack through the wall or ceiling is passed through by a person, radiation or gas, neighbours spying via any 'partition') are common.
- According to ICD patients must either be diagnosed as having delusional disorder or schizophrenia – no separate diagnosis exists for paraphrenia
- Late onset schizophrenia is characterised by (Palmer et al 2001)
    - Fewer negative symptoms
    - Better response to antipsychotics
    - Better neuropsychological performance
    - Greater likelihood of visual hallucinations
    - A lesser likelihood of formal thought disorder
    - A lesser likelihood of affective blunting
    - A greater risk of developing Tardive dyskinesia (The risk of developing Tardive dyskinesia with older antipsychotics is increased in older people by 5-6 times (Kane 1999)
- There is good evidence that the relatives of very-late-onset patients have a lower morbid risk for schizophrenia than the relatives of early-onset schizophrenia patients. The prevalence of schizophrenia (whether early or late in onset) is approximately 7% in siblings and 3% in parents of all probands with late onset
- Premorbid educational, occupational, and psychosocial functioning is less impaired in late-onset than early-onset schizophrenia although many late-onset patients are reported to have had premorbid schizoid or paranoid personality traits.
- Social isolation and sensory deprivation are significantly associated in many studies of late onset psychosis
- Risk factors for late onset psychosis include age related changes in frontal and temporal cortices, cognitive decline, social isolation, sensory deprivation (hearing loss and visual impairment),

polypharmacy, presence of paranoid and schizoid personality traits, precipitating life events (threat or loss), female sex and family history, albeit weaker than younger onset schizophrenia.

Pharmacological Treatment:

- Lewy body dementia must be excluded before antipsychotics are used. Elderly are generally more prone to late-emerging (tardive) extrapyramidal neurological effects than younger adults. Also beware of vascular risks associated with second-generation antipsychotics.
- A Cochrane review failed to identify any eligible studies for the treatment of late onset schizophrenia. Atypical antipsychotics, which have a better side-effect profile, are considered to be more suitable for elderly people. But concerns have been raised regarding the safety of atypical antipsychotics in psychosis due to dementia. The committee on the safety of medicines (CSM) advises that olanzapine and risperidone are associated with a two-fold increase in the risk of stroke in elderly patients especially in people over 80 years. This restriction has been extended to other atypical antipsychotics
- In elderly people, age-related bodily changes affect the pharmacokinetics and pharmacodynamics of antipsychotic drugs, which have numerous side effects that can be more persistent and disabling in older people. So, follow the principle '**START LOW AND GO SLOW'**
- Research literature on the use of conventional antipsychotics suggests significant improvement in psychotic symptoms with the use of haloperidol and trifluoperazine hydrochloride
- The usefulness of clozapine for treatment-resistant early-onset schizophrenia is well-established but concerns about the toxicity and the need for monitoring white cell counts due to more frequent occurrence of agranulocytosis has led to limited use in older patients and should probably be used in treatment resistance and severe tardive dyskinesia

D. Neurotic disorders

- Multiple factors including physical frailty, major life events, bereavement, social isolation, poor self-care and insecure personality may contribute to new onset neurotic symptoms in the elderly. The estimated prevalence of neurotic disorders is between 1-10% with a female predominance. Anxiety disorders are the most prevalent psychiatric disorders, excluding the dementias in people over age 65. The prevalence of anxiety disorders decreases with increasing age.
- The most prevalent anxiety disorder among older adults is phobic disorder. Phobias described by elderly people are similar to those seen in younger adults, although some suggest the fear of falling are more commonly seen in old age
- The least common anxiety disorder in this age group is panic disorder.
- Non-specific anxiety symptoms, hypochondriacal and depressive symptoms predominate. Obsessional, phobic, dissociative and conversion disorders are less common.
- In the elderly, the abuse of sedative drugs and alcohol is a common response to anxiety.
- In cognitively impaired patients disturbed behaviour may be the main presenting feature.

- There is an important association between physical illness and neurotic disorders in old age. Most cases of agoraphobia that develop after the age of 65 years are not induced by panic but arise following alarming experience of physical ill health
- There is an increase in referrals to psychiatry due to decreased acceptance of symptoms by the elderly
- Fluoxetine (GAD, PTSD), citalopram (panic disorder), paroxetine (GAD, PTSD) and venlafaxine (GAD) have been licensed in UK for the treatment of anxiety disorders. Patients need to be warned of a transient increase in anxiety in first 1-2 weeks.

E. Alcohol misuse

- Alcohol use disorders in elderly people are common but under-recognised in the elderly; Older people are likely to encounter alcohol disorders at levels of intake lower than the general population due to the effects of physical and cognitive ageing, pharmacokinetic changes, the increased prevalence of co-morbid illness and interactions with prescribed medication
- The recommended 'safe-drinking' levels of intake for the general population (up to 21 and 14 units per week for men and women respectively) may be inappropriately high for older people. There is a lack of guidance on safe levels of alcohol intake and most guidelines recommend no more than one drink per day for older people.
- Men are more than twice as likely as women to exceed weekly safe-drinking limits). But women report more late onset of alcohol problems. Widowed or divorced men were more likely to engage in heavy drinking. In contrast, among older women those who are married had the highest level of alcohol consumption.
- Alcohol use disorders may be described as being early onset or late onset. In early onset variant, patients have had a life long pattern of problem drinking and have probably been alcoholics for most of their lives. These individuals develop alcohol problems in their 20's or 30's and there is often a family history of alcoholism
- They are more likely to have physical and psychiatric illness. In the late-onset variant, patients first develop drinking problems at 40-50 years of age. They have fewer physical and mental health problems. Often a stressful life event precipitates or exacerbates their drinking. This group is more receptive to treatment and more likely to recover spontaneously from alcoholism.
- Management of older adults with alcoholism should follow similar principles of managing younger adults. Chlordiazepoxide can be used to treat withdrawal symptoms. The three medications used to promote abstinence and reduce relapse are disulfiram, acamprosate and naltrexone. Parenteral or oral thiamine should be given to prevent development of Wernicke-Korsakoff syndrome.
- A number of studies recommend that disulfiram should not be prescribed to elderly people because of the increased risk of serious adverse effects especially precipitation of acute confusional state. It is contraindicated in patients with a history of hypertension, cardiac failure, stroke, or ischemic heart disease and so should be rarely used in older people.

- ✶ **Other substance use disorders:** Illicit drug use in older people is far less of a problem in comparison to alcohol use disorders and medication use disorders. The ECA (Epidemiological Catchment Area) data suggests a lifetime prevalence rate for illegal drug use of only 1.6% for older people. This is substantially lower compared to younger adults and adolescents.
- ✶ Benzodiazepines are the most commonly prescribed psychotropic drugs in older people with one study of community dwelling older people in Ireland demonstrating that 17% of participants were prescribed benzodiazepines. There is a higher risk of falls, confusional state and amnesia in the elderly.

F. Suicide and attempted suicide in old age:

- ✶ **Attempted suicide:** The incidence of deliberate self-harm is highest for the young and declines with age, whereas that for completed suicide rises with age. Suicidal intent behind acts of deliberate self-harm in older people is significantly greater than in younger adults.
- ✶ In clinical practice it is therefore wise to consider deliberate self-harm in those over 75 as failed suicide. Any act of deliberate self-harm suggests depression, as elderly people rarely take manipulated overdoses.
- ✶ As with younger attempters' females outnumber males at a raw number ratio of approximately 3:2 but the proportionate gender ratio is approximately 1:1 because fewer males survive into old age. In cases of completed suicide where men clearly outnumber women
- ✶ Deliberate drug overdose is the most commonly used method for DSH at all age in developed countries; in other countries, corrosive poisons are often used. The most common types of drugs for overdose are benzodiazepines, analgesics and antidepressants. Self-cutting is the next most frequent method of DSH.
- ✶ Older people who self-harm are more likely to be assigned a psychiatric diagnosis after DSH, about half suffering from a major depressive disorder, up to a third from alcohol abuse and under 10% from other disorders. Only about 10% have no psychiatric diagnosis at all. Alcohol abuse together with depressive disorder augments the risk of DSH in older people
- ✶ Risk factors for deliberate self-harm in elderly people include:
  - Physical illness
  - Widowhood and divorce or separation from a co-habitee
  - Social isolation and loneliness
  - Simply living alone
  - Unresolved grief usually after death of a spouse is a commonly found risk factor for DSH.
- ✶ **Completed suicide:** The suicide rates are still highest in the older cohort in most countries. Men outnumber women by about 3 or 4 to 1 in most countries. Suicide at all ages is associated with divorce, widowhood and single marital status. Widowers are more likely to kill themselves than widows, which have relevance to old age psychiatry.
- ✶ In the UK, drug overdose especially in women and frequently with a combination of analgesics, hanging (especially in men), suffocation or jumping from tall structures are the preferred methods.
- ✶ Suicide in older persons is marked by careful planning and about half of the victims leave a note to indicate why or to confirm that they have killed themselves. Suicide acts are generally rare but half

of those that do occur involve people over 65. A previous history of suicide attempt (DSH) is found in about a third of those older people who kill themselves.

- 70% of older suicide victims suffer from major depressive disorder at the time of death. Chronic symptoms of depression and the first depressive illness in later life are associated with a greater risk of suicide. Untreated or inadequately treated depressive illness is also found more commonly in elderly suicides. Co morbid physical illness is a risk factor for suicide in older people.
- Social risk factors for suicide in older people include social isolation, lack of someone to confide in, concerns over dependents or a move from home to residential care. Bereavement by itself is no more of a risk factor in older than younger suicides, but a grief reaction prolonged for more than a year has been found to increase the risk.
- Using *standardized measures of personality traits in the* Monroe County sample, Duberstein and colleagues (1994) demonstrated that suicide in subjects over 50 years of age was associated with higher levels of Neuroticism (N) and lower scores on the Openness to Experience (OTE) factor of the NEO Personality Inventory.
- Harwood and colleagues studied subjects over the age of 60 years and reported that a*nankastic (obsessional) and anxious traits*, were significantly associated with both depression and suicidality in older adults (Harwood et al 2001). Ref: Oxford textbook of psychiatry; 2nd edition; pg 1565-1566

G. Personality disorders in old age

- Roughly 5-10% of older people exhibit features of personality disorder (overall prevalence). Certain traits such as cautiousness and obsessionality and compulsive traits become more prominent in old age. Introversion increases with age. (Howard & Bergmann 1993)
- Early studies found a rate of 4% for community prevalence of 'character disorders' including paranoid states in old age (Kay et al., 1964). Some behaviour such as social withdrawal becomes more likely because of physical or sensory disabilities or undiagnosed illness, including depressive illness. Paranoid traits may intensify, especially in situations where there is increasing social isolation.
- Psychopathy is said to burn-out with advancing age and criminal behaviour is rare in the elderly (approximately 1% of male offenders are >60 yrs).
- A lower prevalence of Cluster B disorders in older adults has been suggested, but a meta-analytic review of 30 studies published between 1980 and 1997 found little difference in the prevalence and cluster distribution of personality disorders between younger (<50 years) and older (>50 years) adults. **(Abrams et al., 1999).** Note that a British study (Cohen, 1994) suggested that older subjects were significantly less likely than younger subjects to have any PD (6.6% vs 10.5%;OR 0.42). This difference was almost entirely attributable to a threefold higher prevalence of dramatic PDs (Cluster B) in the younger cohorts. The largest differences were in antisocial PD and histrionic PD. This study found that the highest prevalence of a PD in elderly was for Obsessive-Compulsive PD (-III) at a rate of 3.3%.
- The diagnostic criteria for personality disorders are somewhat age-biased as they require evidence of dysfunction from younger ages – this information may not be available when diagnosing personality disorders in elderly for the first time

* No specific cluster differences are noted in the prevalence rates of personality disorders in the elderly, though a lower prevalence of cluster B is suggested

Association with other diagnoses:

* Morse and Lynch (2004), using the self rated personality disorder Inventory showed that the patients with PD were four times more likely to have persistent or relapsing symptoms of depression. Both generalised anxiety disorders and substance use disorders were more common in the presence of a personality disorder
* Avoidant, dependent and compulsive traits are particularly likely to occur in patients with a depressive illness, irrespective of age.
* Hypochondriacal personality disorder is associated with psychotic depression.
* Many patients with late paraphrenia have never married and have lived alone for some time suggesting that there may have been personality problems (paranoid personality disorder) predating paraphrenia.
* Personality changes occur in organic disorders (Burns 1992) and negative personality changes are reported in two-thirds of people with dementia. Four patterns of personality change have been reported- alteration at onset of dementia, with little subsequent change; ongoing change with disease progression; regression to previously disturbed behaviours and no change at all.
* **Diogenes syndrome:** Also called ***'senile squalor syndrome'*,** this refers to a syndrome of self-neglect in older people, in which eccentric and reclusive individuals become increasingly isolated and neglect themselves, living in filthy, poor conditions. It can be seen as the response of someone with a particular personality type to the hardships of old age and loneliness (Howard & Bergman 1993). Patients are often oblivious to their condition and resistant to help, necessitating intervention and management is notoriously difficult as it is often impossible to form a therapeutic alliance with the patient. Often but not always-hoarding of possessions often valueless and perceived by others as rubbish (***syllogomania***) encroaching severely on the living space of the home is seen and is regarded as an important feature of this condition

6. Sleep disorder in later life

> ***SLEEP CHANGES IN THE ELDERLY***
>
> *Reduced total sleep time*
> *Increased daytime napping*
> *Increased nighttime arousals and recalled awakenings*
> *Longer sleep latency*
> *Increased stage 1 and 2 sleep*
> *Reduced slow wave sleep*
> *Shorter REM latency*
> *Reduced REM sleep*

- Sleep disorders are common in older people – but should not be regarded as part of normal ageing. Ageing is associated with changes in sleep architecture and altered sleep habits but not necessarily a reduced quality of sleep.
- Sleep disorders are commonly associated with poor mental or physical health but are under recognised; further non pharmacological measures are underused in treatment.
- Insomnia in the elderly is associated with depression, heart disease, pain and memory problems. Polypharmacy can also contribute to sleep problems
- Several types of sleep disorders are common in the elderly: Insomnia, circadian rhythm disorders, Restless legs syndrome (RLS), REM sleep behaviour disorder (RBD), Obstructive Sleep Apnoea (OSA)
- ***Drug induced sleep disorders***: Tricyclic antidepressants can reduce REM sleep, cholinesterase inhibitors increase REM sleep. Both of the above effects are mediated via the cholinergic neurons of the thalamocortical arousal branch (part of the ascending reticular activating system). Dopamine deficiency or antagonism can lead to sleep related movement disorders eg RLS and PLMD via the hypothalamic aminergic arousal branch (also part of the RAS). SSRIs increase slow wave sleep but reduce REM
- ***Insomnia*** is an umbrella term – includes difficulties in initiating and maintaining sleep and subjective complaint of non restorative sleep. It is the commonest sleep disorder in old age; it can significantly impact on quality of life, contributes to illness, institutionalisation and falls.
- Symptoms of insomnia need to persist over 2 weeks and contribute to impaired functioning for interventions to be prescribed. Transient symptoms are common (30-60%) in older adults, particularly in elderly females
- Psychiatric disorders commonly associated with insomnia are mania, depression, OCD, panic disorder and PTSD. Insomnia with sleep fragmentation is particularly common in neurodegenerative disorders e.g. Alzheimer's, DLB and PD. REM sleep behaviour disorder can be an early clinical marker for synucleinopathies (Lewy Body dementia, Multi System Atrophy or Parkinson's)

Treatment of insomnia (derived from http://cks.nice.org.uk/insomnia#!topicsummary)

- Treat underlying cause Advise re: driving (but do not need to inform DVLA)
- Sleep hygiene advice -e.g. https://www.sleepfoundation.org
- Pharmacological treatment options – short acting benzodiazepines e.g. temazepam, Z-drugs, (both work via GABA receptors), melatonin agonists (only if over 55 yrs and symptoms lasting longer than 4 weeks)
- Sedating antidepressants are sometimes used if comorbid depression

* If symptoms still present after 2 weeks refer to IAPT for CBT, or other behaviour therapy (e.g. stimulus control therapy, sleep restriction therapy, relaxation therapy)
* Use of hypnotics should be restricted to patients who meet diagnostic criteria, treatment duration should be short (no longer than 2 weeks)
* **Circadian rhythm disorders:** Particularly common in nursing homes due to inadequate light exposure – secondary to dim light and immobility. Degeneration of the suprachiasmatic nucleus (circadian clock) probably contributes to circadian rhythm disorders. '***Advanced sleep phase syndrome'*** probably commonest rhythm disorder in older adults –fall asleep several hours earlier than conventional norms and wake in the very early morning. **Treatment** - bright light therapy, early evening administration of melatonin, chronotherapy (advancing sleep times gradually each day)
* **Sleep related breathing disorders** include sleep hypopnea and apnoea. Bed partner history is crucial. It is associated with high morbidity and mortality. Polysomnography is needed to confirm the diagnosis. **Treatment** - weight reduction, CPAP therapy, uvulopalatopharyngoplasty (UPPP), oral appliances – modify position of mandible and tongue but less effective than CPAP
* **REM Behaviour disorder (RBD): It is a p**arasomnia, characterised by lack of normal muscle atonia during REM sleep. Enactment of dream activity is noted with increased risk of accident to sleeper and bed partner. Prevalence is much higher in Parkinson's disease (15-34%), MSA (90%) and LBD than in general population (0.5-0.8%). It can predate dementia diagnosis by several years or decades. **Treatment** – safe sleeping environment essential, clonazepam at bedtime, review antidepressants if temporal relationship between use and onset of RBD. Other options include melatonin and pramipexole.

7. Psychosexual disorders in old age

- Age related changes in levels of oestrogen in women may cause vaginal dryness and atrophy, dyspareunia. Decline in testosterone in men after the fifth decade results in decreased sexual desire. Speed and intensity of response to sexual stimulation tends to reduce in both sexes with increasing age. Impotence affects 10-20% of men aged 70 years or over
- Despite of the above, several cross sectional studies have shown that sexual satisfaction does not decline with age.
- It is likely that sexual dysfunction has a high prevalence in older age groups presenting to psychiatric services, but sexual history is rarely covered during clinical assessment. Sexual history should be part of psychiatric history taking.
- Sexual problems in older people are similar to those in younger people – eg erectile dysfunction, vaginismus, anorgasmia, dysparaeunia. Physical and psychological factors frequently combine in cases of sexual dysfunction in the elderly. Medical causes of sexual dysfunction include Parkinson's disease, stroke, arthritis and incontinence. Drugs causing erectile dysfunction include alcohol, benzodiazepines, antidepressants like trazodone (priapism), beta-blockers, antihypertensive like thiazide diuretics and spironolactone.
- Physical illness may result in anxiety around the safety of sexual activity (e.g. after an MI or CVA), undermine self-confidence (e.g. post mastectomy or colostomy), and may have a direct effect in reducing desire. Older people are more likely to undergo surgery, which may have a direct effect on sexual function, e.g. prostatectomy, hysterectomy, genital tumour removal
- **Inappropriate sexual behaviour in dementia** is not particularly common – around 7% overall of people with Alzheimer's dementia, greater in care home settings (approx. 18%). It may be a specific manifestation of frontal lobe pathology or part of a more general disturbance seen in Alzheimer's disease. The reported changes may include inappropriate sexual talk, sexual acting out, implied sexual acts, false sexual allegations. ABC system (antecedents, behaviour and consequences) can be very useful in understanding these behaviours and designing appropriate interventions. Capacity to consent to sexual activity may be a concern in people with dementia, particularly around forming new relationships – important to remember that capacity is decision specific. Of note, adaptation to the role of carer may also adversely affect sexual desire. Loss of interest or an increase in sexual demands can both cause problems.

8. Miscellaneous topics

Psychotherapy in older adults

- There are many indications for the use of psychological treatments in older adults, which mirror the indications in younger people
- Older adults should have access to the full range of psychological treatments, although in reality services may be more difficult to access
- CBT is probably the most commonly used treatment, also can be useful for caregivers of those with dementia
- Cognitive impairment may affect the ability of patients to engage with psychological treatments

- Transference issues in psychotherapy may be altered as the therapist is more likely to be younger than the recipient
- Dependency, loss of sexuality and death can be important factors

Bereavement and adjustment disorders

- Irrespective of age, a third of those who lose a spouse meet criteria for major depression in the first month after the death, and half of these remain clinically depressed 1 year later.
- Treatment should be initiated earlier than this in patients with suicidal ideation, severe functional impairment, a prior history of depression, or other signs of a severe depression.
- In the elderly population, during the first year of bereavement, 10–20% of surviving spouses develop symptoms of depression that require treatment.
- Parkes and later Grace and O'Brien found that bereavement life events were more common in those with early onset depression; this suggests that older people may be better able to cope with bereavements even though they are at increased risk of experiencing them.
- Elderly are less likely to become depressed than younger adults during the first months of widowhood. But rates of major depression are similar in all ages by the end of the second year.
- The prevalence of major depression continues to increase during the second year of bereavement, by the end of which 14% of bereaved individuals have major depression, compared with 1–4% of the elderly population in general (Alexopoulos, 2005)

Grief Reactions
Normal grief reaction **Phase 1** - Shock and protest – includes numbness, disbelief and acute dysphoria **Phase 2** Preoccupation – includes yearning searching and anger **Phase 3 –** disorganization – includes despair and acceptance of loss **Phase 4 –** resolution
In normal grief reactions substantial improvement is expected within 2 months to 6 months, and those who continue to meet criteria for major depression after this time period should receive antidepressant or psychotherapy
Abnormal grief
Inhibited grief – absence of expected grief symptoms at any stage **Delayed grief –** avoidance of painful symptoms within 2 weeks of loss **Chronic grief –** continued significant grief related symptoms 6 months after loss

Notes prepared using excerpts from

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